

Fragmentation and how it affects performance

When we create a file, it takes some space on the disk and when we delete it completely (even from the Recycle Bin), the space that the file was sitting on is marked as empty. When we do this frequently (which is a given in today's day and age), logical gaps appear where files used to be previously. When a new file is added but there is no empty space large enough to accommodate the whole file, the OS splits the file into multiple pieces and stores them wherever it can find empty space. It links the pieces together with markers which indicates where the next chunk of the file is present. This is called fragmentation.

Fragmentation decreases the performance of an HDD by increasing the time it takes to move the read/write head to go to different chunks to read the file. When you defragment the file, the OS rearranges files on disk in a manner that all files are allocated in continuous (contiguous in file system terminology) spaces. Now you know why you should defragment your disk from time to time.

SSDs

SSDs or Solid State Disks behave like RAM in terms of data access methodology. The pieces of data are stored in whatever space is available and can be accessed at the same speed no matter what part of the storage space they're located on. Much like how USB pendrives behave, only faster.

SSDs are typically very energy efficient because they do not have moving parts. This eliminates the need to power a motor for spinning the disks. They are also not affected by fragmentation because of negligible seek times – after all, there are no heads to be moved.

What storage device should I use for my PC?

For the most part, HDDs should suffice. However, if you measure time in milliseconds instead of minutes, then an SSD might be more your calibre.

Basic Rig

The Toshiba 1 TB DT01ACA100 with 1 TB storage @ 7200 RPM and a 32 MB cache on board delivers a decent performance. We do think 1 TB is essential to a digital life, especially when it comes at about ₹3100.